

HS4002: Lecture 6B

Multivariate Analysis II

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Table of Contents

1. Re-cap.
2. Linear Probability Model
3. Logistic Regression
4. Math Behind the Logistic Model
5. Math Behind Estimation
6. Interpreting Results
7. Comparing Models

Re-cap.

For the midterm:

- Two-hour long session, I estimate you should be able to complete it in an hour though.
- Bring paper and stationery with you.
- Bring a calculator if you want, but there's no real taxing calculation that you can't do by hand.

How do we interpret regression output?

1. Note how the dependent (outcome) variable is measured.
→ How do you interpret changes in the outcome variable?
2. Note how independent (predictor) variables are measured.
→ How is a “one unit” change in the independent variable interpreted?
3. Note the coefficient estimates.
4. Note the p-value of the coefficient estimates.

Interpreting coefficient estimates

Let's assume a model where the dependent variable is a continuous measure of income.

- If independent variable is a continuous variable (e.g. years of education):
 - "We observe that on average a one unit change in years of education is associated with a XXX change in income ($p < 0.01$), ceteris paribus."

Interpreting coefficient estimates

If independent variable is a categorical variables (e.g. race or gender):

- Note the reference group (what's coded as 0).
 - i.e. if gender is coded as men = 0, and women = 1, then men are the reference group.
- "We observe that on average being Malay is associated with a XXX change in income when compared to the reference group (Chinese) ($p < 0.01$), ceteris paribus."

Interpreting coefficient estimates

We are often interested in second-order interaction effects apart from the main effect, e.g. gendered effects on education on income.

- For interaction of variables (e.g. race $\times$ years of education):
 - "Apart from the main effects of race and education, we also observe an interaction effect between race and education. For Malays, each year of education is associated with a decreased income of \$XXX when compared to Chinese ($p = 0.03$)."

- If null effect:
 - "We observe that no associations between age and income ($p = 0.47$) , ceteris paribus."

Linear Probability Model

Binary Outcomes?

What happens when the dependent variable is a binary variable?

The Linear Probability Model

We can continue to use the OLS:

$$P(y = 1|x_1, x_2 \dots x_k) = \beta_0 + \beta_1 x_1 + \dots \beta_k x_k + \epsilon$$

where we can interpret the beta coefficients to reflect changes in the probability of success ($y = 1$) when there's a change in independent variable, *ceteris paribus*.

How to interpret results

If our dependent variable was marriage, we could have $y = 1$ to be married and $y = 0$ to be not married.

- β_{income} would tell us the expected change in the probability of marriage when X_{income} increases by one unit, ceteris paribus.

Limitations with the LPM

- If we plug certain combinations of values for the independent variables into, we can get predictions either less than zero or greater than one.
- Because of this, the convention within sociology is to often turn to other models.

Logistic Regression

The logistic regression is an example of a binary response model,

$$P(y = 1|x_1...x_k) = G(\beta_o + \beta_1x_1 + ... \beta_kx_k + \epsilon)$$

where G is a function that takes on values strictly between 0 and 1.

Math Behind the Logistic Model

In the logistic (or, logit) model, G is the logistic function such that

$$\begin{aligned} G(\beta_o + \beta_1 x_1 + \dots \beta_k x_k + \epsilon) &= \frac{\exp(\beta_o + \beta_1 x_1 + \dots \beta_k x_k + \epsilon)}{1 + \exp(\beta_o + \beta_1 x_1 + \dots \beta_k x_k + \epsilon)} \\ &= \frac{1}{1 + \exp[-(\beta_o + \beta_1 x_1 + \dots \beta_k x_k + \epsilon)]} \end{aligned}$$

where G is a function that takes on values strictly between 0 and 1.

$$P(y = 1|x_1...x_k) = \frac{1}{1 + \exp[-(\beta_o + \beta_1x_1 + ... \beta_kx_k + \epsilon)]}$$

$$\frac{P(y = 1|x_1...x_k)}{P(y = 0|x_1...x_k)} = \frac{P(y = 1|x_1...x_k)}{1 - P(y = 1|x_1...x_k)}$$

...

$$= \exp(\beta_o + \beta_1 x_1 + ... \beta_k x_k + \epsilon)$$

Taking the log of this gives us ...

$$\log \left[\frac{P(y = 1|x_1 \dots x_k)}{P(y = 0|x_1 \dots x_k)} \right] = \beta_o + \beta_1 x_1 + \dots \beta_k x_k + \epsilon$$

The term on the left, $\log \left[\frac{P(y=1|x_1 \dots x_k)}{P(y=0|x_1 \dots x_k)} \right]$, are the log odds of $y = 1$.

Math Behind Estimation

We can express the density of y_i given $\mathbf{x}_i$ as

$$f(y|\mathbf{x}_i; \beta) = [G(\mathbf{x}_i; \beta)]^y [1 - G(\mathbf{x}_i; \beta)]^{1-y},$$

where y is a binary variable, $\mathbf{x}_i$ is a vector of independent variables for observation i , and β is the vector of β coefficients.

Then the log-likelihood function for observation i can be expressed as

$$\ell_i(\beta) = y_i \log[G(\mathbf{x}_i; \beta)] + (1 - y_i) \log[1 - G(\mathbf{x}_i; \beta)].$$

- We use maximum likelihood estimation to produce estimates of $\hat{\beta}$.
- The MLE estimate of β is the set of values that maximizes this log-likelihood for a sample of size n .

Interpreting Results

- It's easy to interpret how changes in the independent variables affect changes in log odds.
 - You do it the same way as you do a linear model.
 - e.g. "A one unit change in X is expected to cause an increase in the log odds of Y by [number] ($p < 0.01$), ceteris paribus."
- You can also produce an interpretation of the odds by taking the exponential on both sides of the equation earlier.
 - e.g. "A one unit change in X is expected to cause the odds of Y to change by a factor of $\exp(\hat{\beta}_X)$ ($p < 0.01$), ceteris paribus."

- Note that they DO NOT indicate the magnitude of the change in the probability of the outcome.
- Partial effects on the probability of the outcome are hard to summarize because the scale factor depends on the values of the explanatory variables.
 - i.e. there is a non-linear relationship between the probability of interest, and the independent variables in the model.

- “Marginal effects” measure how a small change in an independent variable X influences the dependent variable.
- Reminder the derivative from calculus?
→ $\frac{d(y)}{d(x)}$ gave you the gradient of a graph, for example.
- Many scholars like to use marginal effects to interpret models with more difficult functional forms, such as the logistic regression, Poisson regression, and so on.

- Interpretation depends on whether X is a continuous or discrete variable.
 - If discrete (e.g. binary variable), it is the expected change in the predicted probability when an independent variable X increases by one unit, *ceteris paribus*.
 - If continuous, it is the expected change in the predicted probability when an independent variable X increases by a small ('infinitesimal') amount, *ceteris paribus*.

- Different types of marginal effects
 - **Average marginal effect (AME)**: averages marginal effects across all observations in the data.
 - **Marginal effects at the mean (MEM)**: estimates marginal effects at the average values of the independent variables.

Comparing Models

- We can use the same AIC statistic as with OLS regression.
 - We interpret it the same way, where a smaller AIC value indicates a better model when comparing two nested models.
- There are also pseudo R^2 values that can be interpreted the same way as R^2 values previously encountered.

- We can use Likelihood-Ratio Tests to compare a restricted model with a unrestricted model.
 - Recall that the restricted model is the one with the restrictions required under the null (aka the "smaller" model).
- This is analogous to the F-test that we did last week for linear models.

Likelihood-Ratio Test

Let LR be the likelihood ratio statistic,

$$LR = -2 \ln \frac{\mathcal{L}_{ur}}{\mathcal{L}_r}$$

where $\mathcal{L}_{ur}$ is the likelihood of the unrestricted model (the one with more parameters) and $\mathcal{L}_r$ is the likelihood of the restricted model (the one with fewer parameters).

Likelihood-Ratio Test

- The null hypothesis is that the smaller, "restricted," model is the TRUE model.
- LR has an approximate chi-square distribution under the null.
- If p is sufficiently small, we are inclined to reject the null and conclude that the restriction conditions is unlikely to be true (i.e. the restricted parameters do not have coefficients of zero.)

Thanks for listening :)